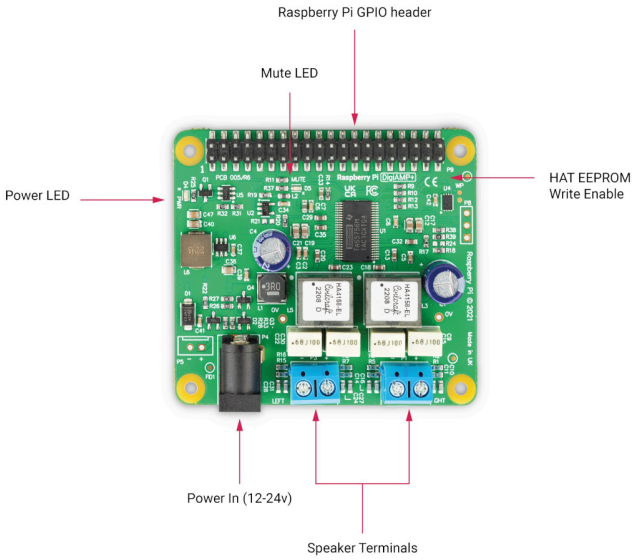
Raspberry Pi DigiAMP+

Source: https://github.com/raspberrypi/documentation/blob/7376533ab86ea07acc77ef6a9f3c3e57fcff833d/documentation/asciidoc/accessories/audio/digiamp_plus.adoc (CC BY-SA 4.0; adapted for this benchmark)
Image source: https://raw.githubusercontent.com/raspberrypi/documentation/7376533ab86ea07acc77ef6a9f3c3e57fcff833d/documentation/asciidoc/accessories/audio/images/DigiAMP%2B_Board_Diagram.jpg (CC BY-SA 4.0)



Raspberry Pi DigiAMP+

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With Raspberry Pi DigiAMP+, you can connect 2 passive stereo speakers up to 35 W with variable output, making it ideal for use in Raspberry Pi-based hi-fi systems.

DigiAMP+ uses the Texas Instruments TAS5756M PowerDAC and must be powered from an external supply.

It requires a 12-24V DC power source (the XP Power VEC65US19 power supply is recommended).

NOTE: The DigiAMP+ is designed to work with Raspberry Pi 3 and earlier devices. If you use it with a Raspberry Pi 4 or later devices, use an externally-powered USB hub to prevent the USB ports from adding extra load to the device. Failure to do so is likely to result in a low power warning.

DigiAMP+'s power in barrel connector is 5.5 mm × 2.5 mm.

At power-on, the amplifier is muted by default (the mute LED is illuminated). Software is responsible for the mute state and LED control (Raspberry Pi GPIO22).

DigiAMP+ is designed to provide power to the Raspberry Pi and DigiAMP+ together in parallel, delivering 5.1V at 2.5amp to the Raspberry Pi through the GPIO header.

WARNING: Don't apply power to the Raspberry Pi device's own power input when using DigiAMP+.

Pinouts

P5 | Alternative power input for hard wired installations (polarity must be observed).

P8 | TAS5756m Internal GPIO1/2/3

Speaker Terminal Polarity

Polarity is printed on the silk screen above the speaker terminals.

Looking down at the top of the board, and with the power connector on the left, both terminals have the same polarity: "-" on the left and "+" on the right.